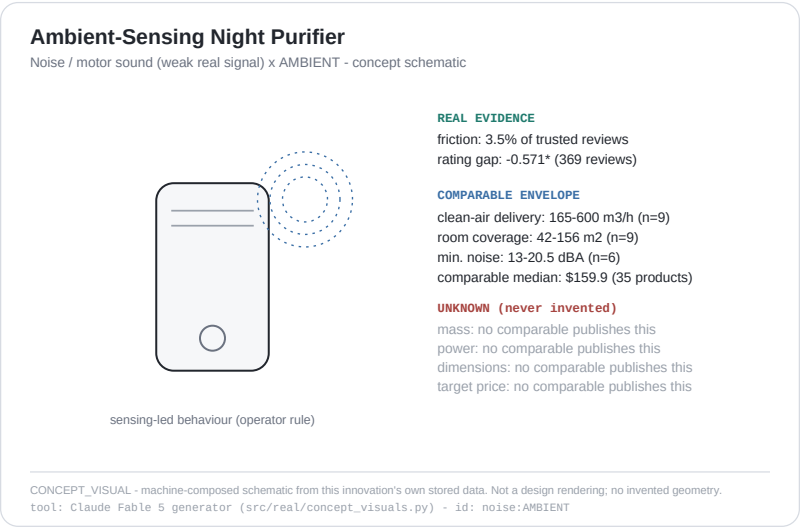


# Ambient-Sensing Night Purifier

Innovation dossier - 1. The idea

PROPOSITION

AMBIENT applied to the real 'Noise / motor sound (weak real signal)' friction: Optimize for hours-actually-run under real noise tolerance, not peak lab CADR. Assumption challenged: "Clean-air delivery rate (CADR) is the primary performance basis." -> What if performance were reported as a loss-rate curve or realized exposure reduction, not a lab CADR number?



TARGET USER / CONTEXT

Households whose real reviews carry this friction: 369 reviews across 77 real products (69.6% verified purchase, 2004-10-14-2023-08-08).

UNKNOWN - no persona/behavioural dataset exists in this pipeline; only the review evidence above is real.

MECHANISM - METHOD CHOICE

**AMBIENT - Explicit interaction -> invisible environmental behaviour.**

METHOD\_CHOICE - authored design vocabulary

PRODUCT ARCHETYPE - DESIGN RULE

**connected:** Likely - operator AMBIENT (invisible environmental behaviour) requires sensing.

room scale, mobility, size class, power class, noise ambition, maintenance burden, filter service dependency: UNKNOWN - no real data in this pipeline for this field.

DESIGN\_RULE - each populated field follows from the operator's own fixed definition, never from evidence

Figure: CONCEPT\_VISUAL

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT\_VISUAL, never a digital model or physical prototype.

artifact visual-noise:AMBIENT - /concept-visuals/noise\_AMBIENT.svg

ENGINEERING ENVELOPE

comparable basis: 10 verified official Versuni/Philips air-purifier families (room-scale)

| metric                     | value from verified comparables                                                                              | epistemic type         |
|----------------------------|--------------------------------------------------------------------------------------------------------------|------------------------|
| performance cadr m3h       | 165-600 m3/h (n=9 comparables)                                                                               | OBSERVED_COMPARABLE    |
| room coverage m2           | 42-156 m2 (n=9 comparables)                                                                                  | OBSERVED_COMPARABLE    |
| acoustic min dba           | 13-20.5 dBA (n=6 comparables)                                                                                | OBSERVED_COMPARABLE    |
| target mass kg             | No verified comparable publishes this value - left unknown, never estimated.                                 | UNKNOWN                |
| target power w             | No verified comparable publishes this value - left unknown, never estimated.                                 | UNKNOWN                |
| target dimensions          | No verified comparable publishes this value - left unknown, never estimated.                                 | UNKNOWN                |
| reference market price usd | median \$159.9 (range \$28.98-\$995.0, n=35) - market price of affected products today, not a proposed price | REFERENCE_MARKET_PRICE |

State: DEVELOPING

Non-dominated and grounded in a real research tension.

WHY NOW - THE OBSERVED REALITY

3.5% of trusted reviews (369 reviews across 77 real products) carry this friction, with a -0.571\* average rating gap. Signal: Motor/fan noise is a recurring complaint, now backed by peer-reviewed evidence it drives real under-use.

## THE DERIVATION - THREE PARTS, EACH LABELLED

**1. Reality** **OBSERVED**

3.5% of trusted reviews (369 reviews across 77 real products) carry this friction, with a -0.571\* average rating gap. Signal: Motor/fan noise is a recurring complaint, now backed by peer-reviewed evidence it drives real under-use.

**2. Transformation** **METHOD CHOICE**

METHOD CHOICE, not evidence: the authored operator table assigns AMBIENT - "Explicit interaction -> invisible environmental behaviour." - to this friction. Analyst design judgment, applied deterministically.

**3. Product consequence** **RESEARCH TENSION**

Optimize for hours-actually-run under real noise tolerance, not peak lab CADR. Assumption challenged: "Clean-air delivery rate (CADR) is the primary performance basis." -> What if performance were reported as a loss-rate curve or realized exposure reduction, not a lab CADR number?

## PARENT PATHS

**tension:T1 - Filtration vs. noise-driven adherence** **TENSION**

test: Falsified when one design point achieves RP-01's measured primary-room particulate reduction at a sound level that removes noise as a named reason for running the device less (RP-05) - both halves are stored, quantified findings.

**tension:T2 - Constant vs. automatic operation** **TENSION**

test: Falsified when an auto-mode arm reaches the lower bound of RP-07's constant-mode reduction interval while keeping auto's runtime/noise advantage - the two stored confidence intervals would then overlap and the trade-off dissolves.

**tension:T3 - CADR theory vs. real-home performance** **TENSION**

test: Falsified when a CADR-derived sizing prediction matches a source-strength-adjusted measured loss rate within the fan-speed band RP-06 reports, in a home with the central air-handling configuration RP-01 identifies as decisive.

**tension:T6 - Pollutant reduction vs. clinical outcome certainty** **TENSION**

test: Resolved when an independent replication of RP-08 reproduces its clinical effect in a cohort whose particulate reduction matches RP-03's measured range - RP-08's own record names independent replication as the missing step. A failed replication resolves it the other way.

**assumption:A5 - Clean-air delivery rate (CADR) is the primary performance basis.**

**ASSUMPTION TO TEST**

test: Challenged when a certification body in the trend corpus adopts a loss-rate or realized-exposure metric alongside CADR - concretely, when TC-R03 (AHAM Verifide, the one record whose theme tags contain 'cadr': ['certification', 'cadr', 'iaq\_standards']) stops centring on it, or TC-R02 (ENERGY STAR, themes ['regulation', 'energy\_efficiency', 'iaq\_standards'], CADR present only in its prose scope note) adds a non-CADR performance theme tag. Fragile baseline: both documents carry published\_date null, so change detection has no dated anchor.

**assumption:A6 - Continuous, constant operation is the default expectation.**

**ASSUMPTION TO TEST**

test: Challenged when connected+adaptive share exceeds 25% of the corpus (currently 23/237 = 9.7%), or when an independent trial reproduces RP-07's constant-mode filtration under auto operation.

**evidence ids:** taxonomy:noise, RP-05, RP-07, TC-R11

## FIELD EVIDENCE BEHIND THESE PATHS (TITLES)

- Self-reported health impacts of do-it-yourself air cleaner use in a smoke-impacted community
- Real-World Effectiveness of Portable Air Cleaners in Reducing Home Particulate Matter Concentrations
- Reduction of personal PM2.5 exposure via indoor air filtration systems in Detroit: an intervention study
- The impact of portable air cleaners on indoor particulate matter concentrations and perceptions of indoor air quality: a randomized crossover trial in three multifamily buildings
- Analysis of sources and sinks of indoor particulate matter reveals insights into the real-world efficacy of portable air cleaners in a randomized intervention trial
- Indoor environmental health and asthma relief: findings from a longitudinal HEPA filter trial
- Effectiveness of portable HEPA air cleaners on reducing indoor PM2.5 and NH3 in an agricultural cohort of children with asthma: A randomized intervention trial

## DESIGN DNA

**F consumer friction (taxonomy:noise)** **PRESENT**

Real Amazon review-text friction theme (see Consumer Pain methodology).

**O design operator** **METHOD CHOICE**

Authored operator vocabulary applied deterministically (METHOD\_CHOICE, not evidence) - see OPERATORS/THEME\_OPERATORS in src/real/magic\_box\_real.py.

**S signal (noise)** **PRESENT**

Motor/fan noise is a recurring complaint, now backed by peer-reviewed evidence it drives real under-use.

**T scientific tension** **PRESENT**

Filtration vs. noise-driven adherence; Constant vs. automatic operation; CADR theory vs. real-home performance; Pollutant reduction vs. clinical outcome certainty

**R rival gap** **PRESENT**

11 named real competitors measurably weaker on this theme. [Dyson, Plixio, Oreck, Envion, Wynd, Molekule, AMEIFU, AZEUS, AIRTHEREAL, Egofine, Elechomes]

**C versuni capability** **MISSING UNVERIFIED**

No real Versuni internal capability/org-readiness dataset exists in this pipeline.

**A category assumption** **PRESENT**

Clean-air delivery rate (CADR) is the primary performance basis.; Continuous, constant operation is the default expectation.

**E economic condition** **PRESENT**

Real price-weighted exposure computed for this theme (\$84,015.62).

## ECONOMICS - RELATIVE INDICATORS ONLY

**price-weighted exposure: \$84015.62**

SUM of real observed listed prices across affected reviews with a known price. This is a RELATIVE exposure indicator, not a revenue or market-size estimate - it has no units-sold, conversion-rate, or time-period basis. It answers 'which friction touches more expensive products' not 'how much money is at stake per year.'

**comparable market median: \$159.9 (n=35 products) - not this concept's price**

MEDIAN real listed price across the 35 distinct real products affected by this friction that have a known price. This is what products in this segment actually cost today - not a proposed price for a new concept, which this evidence cannot establish.

**category assumptions:** A5, A6

## UNCERTAINTIES

- No real acoustic-engineering trend document exists in this corpus at all - the medium rating is a judgment call made in the absence of any external precedent, not a citation.

## CONTRADICTIONS

- none recorded for this innovation

## OBSERVED VS ESTIMATED VS UNKNOWN

**observed:** friction stats; envelope: performance cadr m3h, room coverage m2, acoustic min dba; DNA F/S/T/R/A/E

**method choice:** operator, archetype design rule; DNA O

**unknown:** envelope: target mass kg, target power w, target dimensions; persona; DNA C

Where this idea stands: DEVELOPING

Non-dominated and grounded in a real research tension.

PROTOTYPE STATE

CONCEPT\_VISUAL

CONCEPT\_VISUAL means a schematic exists, nothing more: no digital model, no physical prototype, no test data.

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT\_VISUAL, never a digital model or physical prototype.

NEXT EXPERIMENT

A real acoustic-engineering standard or published low-RPM motor precedent, if found, would raise this to high; a real internal R&D estimate showing it requires novel hardware would lower it to low.

KILL CRITERION

No authored kill criterion yet - a non-formal innovation earns one by displacing the formal recommendation (Scenario lens) or by a human decision to invest.

CRITIC DIMENSIONS

|                                |                |                                                                                                            |
|--------------------------------|----------------|------------------------------------------------------------------------------------------------------------|
| EVIDENCE                       | SURVIVE        | Real Consumer Pain evidence: prevalence 3.5%, average rating gap -0.571 - materially above the gate floor. |
| ECONOMIC                       | SURVIVE        | Real price-weighted exposure (\$84,016) is at or above the corpus median (\$16,113).                       |
| COMPETITIVE                    | SURVIVE        | Real white space: 11 named real rivals are measurably weaker on this theme.                                |
| HUMAN                          | NEEDS_EVIDENCE | No real usability/human-factors evidence exists in this pipeline for this concept.                         |
| PHYSICAL                       | NEEDS_EVIDENCE | No real lab/engineering feasibility data exists in this pipeline for this concept.                         |
| VERSUNI_FIT                    | NEEDS_EVIDENCE | No real organizational-capability assessment exists in this pipeline for this concept.                     |
| TIMING                         | NEEDS_EVIDENCE | No real market-timing/window evidence exists in this pipeline for this concept.                            |
| ROBUSTNESS                     | NEEDS_EVIDENCE | No real sensitivity/robustness study exists in this pipeline for this concept.                             |
| critic overall: NEEDS_EVIDENCE |                |                                                                                                            |

CRITERIA RESULTS

NEEDS\_EVIDENCE: 32 - PASS: 13 - CHALLENGE: 4 - N/A: 3

|    |           |                                                                                                                                                                    |
|----|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| V8 | CHALLENGE | Operator AMBIENT may add surface area to manage - unverified.                                                                                                      |
| B5 | CHALLENGE | Filtration vs. noise-driven adherence; Constant vs. automatic operation; CADR theory vs. real-home performance; Pollutant reduction vs. clinical outcome certainty |
| P2 | CHALLENGE | Filtration vs. noise-driven adherence; Constant vs. automatic operation; CADR theory vs. real-home performance; Pollutant reduction vs. clinical outcome certainty |
| F3 | CHALLENGE | medium                                                                                                                                                             |

LINEAGE / PROVENANCE

generated by: src/real/innovations\_real.py  
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funnel snapshot: 73aa128628a9304d744c80cb718ecf82ed99b270b6e5d457a0198b59b6041e37  
figure source: /concept-visuals/noise\_AMBIENT.svg (generated 2026-08-31T12:16:37.701217+00:00 by Claude Fable 5 generator - src/real/concept\_visuals.py)  
data source: data/processed/innovations\_real.json + data/processed/funnel\_real.json (deterministic; no invented fields)  
dossier generator: src/real/generate\_innovation\_dossiers.py